AERIS: Attention-Entropy Residue Smoothing for Lightweight Post-Prediction Protein Structure Refinement

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Abstract. Modern protein structure predictors provide strong initial protein models, but downstream workflows can still benefit from lightweight post-prediction refinement. We introduce AERIS, an Attention-Entropy Residue Smoothing algorithm that uses residue-level attention as a refinement prior. Given an input protein structure, AERIS extracts the corresponding sequence, obtains residue representations and attention weights from an attention-bearing protein model, indexes residues by embedding norm and attention entropy, fixes high-importance low-entropy residues as anchors, and applies a small centroid-directed C-alpha smoothing update to the remaining residues. In a pilot benchmark of 19 diverse proteins initialized with Boltz-2 and indexed with ESM-2, AERIS reduced mean C-alpha RMSD from 18.07 to 17.09 Angstroms, corresponding to a mean absolute reduction of 0.98 Angstrom and a mean percentage reduction of 5.94%, with all 19 unique targets showing RMSD improvements compared to ground-truth structures obtained from UniProt. The result supports the feasibility of attention-entropy-guided geometric smoothing to enhance foundational protein structure prediction models.
Keywords: protein structure refinement; protein language models; attention entropy; ESM-2; Boltz-2; RMSD

1. Introduction

Protein structure prediction has been transformed by deep learning systems such as AlphaFold2 and Boltz-2, which demonstrated near-experimental accuracy for many targets and established learned structural priors as central tools in computational biology [1]. More recent biomolecular foundation models extend this direction to complexes and interaction-aware prediction [2,3]. However, prediction quality can still vary across targets, reference coverage, domain organization, and downstream use cases. For that reason, post-prediction refinement remains useful when the goal is to improve a given structure rather than replace the predictor itself.

AERIS is motivated by a simple observation: many modern protein language models contain residue-level attention maps or attention-like pairwise signals. These signals are not limited to one architecture. In the present benchmark, Boltz-2 is used only as an initial structure generator and ESM-2 is used only as an attention source. Conceptually, AERIS can be paired with any structure predictor that outputs coordinates and any sequence or structure model that provides residue-residue attention or a comparable residue-wise saliency signal.

The central hypothesis is that attention entropy can help distinguish residues whose representation is concentrated around specific partners from residues whose attention is diffuse across the sequence. Low attention entropy, especially when combined with a high representation norm, is treated as evidence for a structurally informative residue. AERIS uses these residues as anchors and performs a small C-alpha-level smoothing step on the remaining residues. Therefore, AERIS introduces a predictor-agnostic, attention-source-agnostic refinement algorithm to enhance protein structure prediction models.

2. Related work and positioning

AlphaFold2 established that neural structure predictors can reach high accuracy across many proteins [1]. Boltz models extend biomolecular interaction modeling in an open framework; Boltz-2 is described by its developers as jointly modeling complex structures and binding affinities [2,3]. In this work, Boltz-2 is not treated as essential to the method. It is simply the initial predictor used in the pilot benchmark.

Protein language models provide residue-level representations learned from large protein sequence corpora. ESM-2 and ESMFold demonstrated that atomic-level structure prediction can be projected from

large protein language model representations [4]. The ESM implementation also exposes attention maps, and prior work on transformer protein language models showed that attention maps can encode structural contacts [5]. These observations motivate attention as a source of residue-level structural information.

Although recent protein structure benchmarking recommendations emphasize that local and contact-based measures such as lDDT and CAD-score may be preferable to RMSD in many automated evaluation settings, [6] this study intentionally uses RMSD as a simple, classical, and interpretable metric for algorithmic comparison. AERIS is therefore presented as a proof-of-concept refinement algorithm study focused on measuring whether the proposed attention-entropy smoothing step produces a consistent geometric improvement, rather than as a complete validation of physical model quality.

Contributions

- A predictor-agnostic post-processing formulation: AERIS assumes only an input structure and a residue attention source.
- An attention-entropy residue-indexing rule that combines representation norm with attention entropy to define structural anchors.
- A lightweight centroid-based C-alpha smoothing update that preserves selected anchors while adjusting non-anchor coordinates.
- A pilot evaluation showing consistent RMSD reduction across 19 unique proteins under the current experimental protocol.

3. Method

3.1 Problem setup

Given an input protein structure X with C-alpha coordinates c_i for residues i = 1, ..., L, AERIS produces a refined C-alpha coordinate set X′. The method assumes access to an attention-bearing model M that returns contextual residue embeddings h_i and attention weights A_{ij}. In the pilot implementation, X is produced by Boltz-2 and M is ESM-2. These choices are replaceable: the same refinement rule could be applied to structures from AlphaFold-like, RoseTTAFold-like, Boltz-like, or experimental starting models, provided residue indexing between sequence and structure is available.

3.2 Residue indexing by attention entropy

For each residue i, AERIS computes an embedding-norm score $r_i = \|h_i\|_2$ and an attention entropy score

$H_i = -\sum_j \bar{A}_{ij} \log(\bar{A}_{ij} + \varepsilon)$, where $\bar{A}$ is the attention matrix averaged across available layers and heads. In the current implementation, ESM-2 layer 12 representations are used for h_i, and attention maps are averaged before entropy calculation.

Functional anchors are selected as residues whose embedding norm lies above the 80th percentile and whose attention entropy lies below the 20th percentile. In set notation, $A = \{i : r_i > Q_{0.80}(r) \text{ and } H_i < Q_{0.20}(H)\}$. High-entropy residues are also recorded as flexible candidates, $F = \{i : H_i > Q_{0.80}(H)\}$, but the present coordinate update uses only the low-entropy functional anchors as fixed points.

High-norm residue representations are treated as stronger model-derived signals of structural or contextual importance. Using a percentile threshold gives a simple, scale-independent way to retain only the most informative residues within each protein, rather than relying on an absolute score that may vary across sequences.

3.3 Centroid-indexed C-alpha smoothing

Let μ_A be the centroid of the C-alpha coordinates of the selected anchor residues. AERIS preserves anchor coordinates and moves every non-anchor C-alpha coordinate a small fraction α, the smoothing factor, toward this centroid:

$c_i' = c_i, \quad i \in A$
$c_i' = c_i + \alpha(\mu_A - c_i), \quad i \notin A$

3.4 AERIS refinement algorithm

Input: predicted structure X, attention-bearing model M, percentiles p_norm = 80, p_entropy = 20, step α = 0.2

1. Extract sequence and C-alpha coordinates from X.
2. Run M on the sequence to obtain residue embeddings h_i and attention maps A_{ij}.
3. Compute $r_i = \|h_i\|_2$ and $H_i = -\sum_j \bar{A}_{ij} \log(\bar{A}_{ij} + \varepsilon)$.
4. Select anchors $A = \{i : r_i > Q_{0.80}(r) \text{ and } H_i < Q_{0.20}(H)\}$.
5. Compute anchor centroid μ_A from the C-alpha coordinates of residues in A.
6. For each residue i, keep c_i fixed if $i \in A$; otherwise set $c_i = c_i + \alpha(\mu_A - c_i)$.

Output: refined C-alpha coordinate set X′.

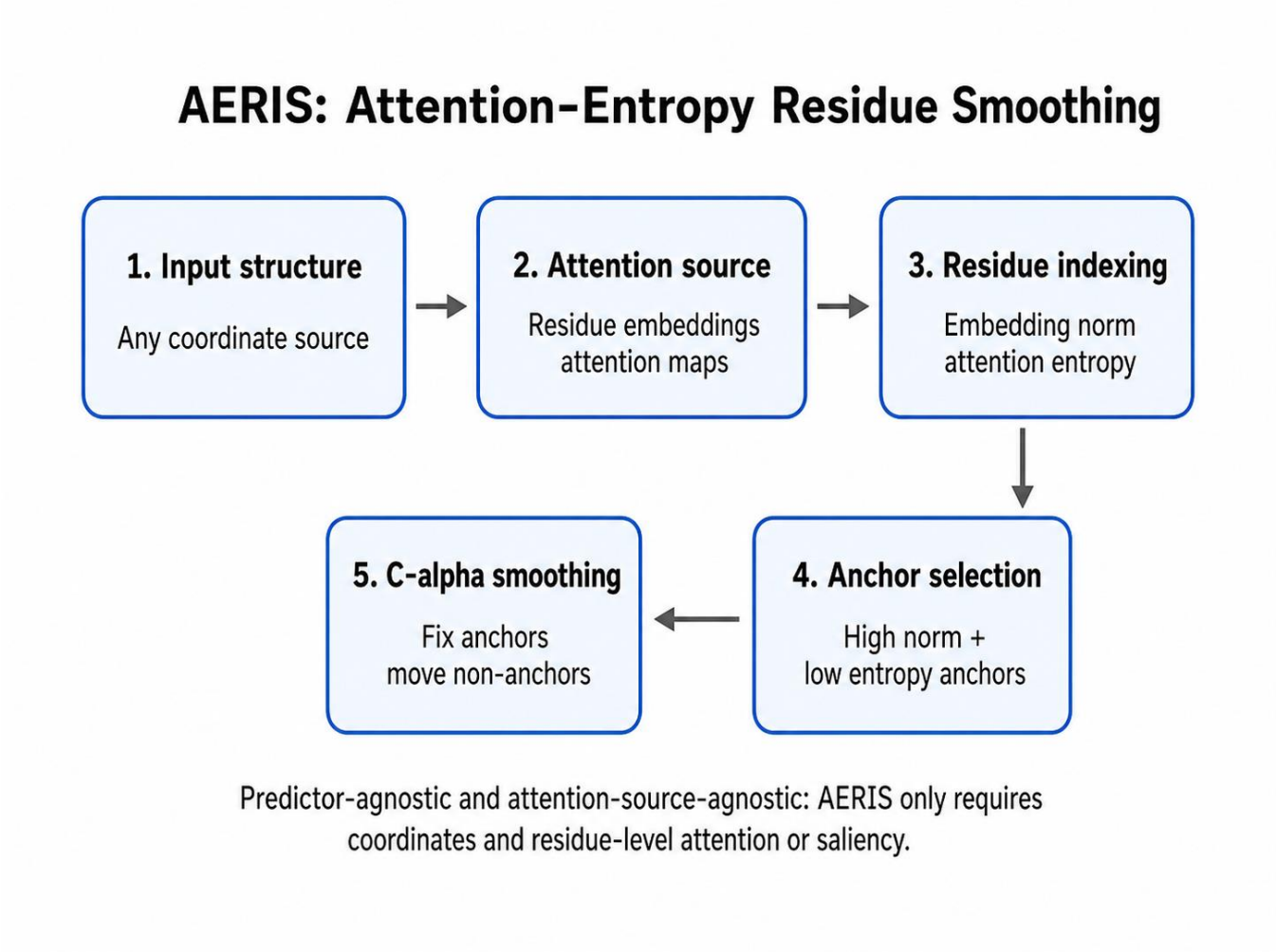

Figure 1. Overview of AERIS. The method takes an input protein structure, obtains residue attention from an attention-bearing protein model, indexes residues using embedding norm and attention entropy, fixes high-importance low-entropy anchors, and applies a small centroid-directed smoothing step to non-anchor C-alpha coordinates.

4. Experimental setup

The benchmark used UniProt IDs as sequence inputs. The benchmark targets were selected as well-studied proteins spanning diverse biological functions. This choice was intended to provide a check of whether AERIS produces consistent geometric refinement across varied protein contexts, including proteins with different sizes, folds, and functional roles.

For each target, a single-chain YAML input was generated and predicted with Boltz-2 using GPU acceleration, use_potentials enabled, one diffusion sample, three recycling steps, and fifty sampling steps. ESM-2_t33_650M_UR50D was then used to extract residue representations and attention maps. Functional and flexible percentile thresholds were both set to 80, corresponding to high-norm residue filtering and low-entropy anchor selection as described above.

Boltz-2 and ESM-2 are used as convenient representative components rather than fixed requirements of the method. Boltz2 provides the initial input structure, while ESM-2 provides residue-level embeddings and attention maps for anchor selection. In principle, AERIS can be applied to structures generated by any predictor, and its residue-indexing step can use any protein language model or attention-bearing model that provides residue-level attention or comparable saliency signals.

Reference structures were obtained by querying UniProt cross-references and downloading the first available PDB entry as ground-truth structures for evaluation. C-alpha RMSD was calculated by extracting C-alpha atoms from prediction, refined, and reference structures, truncating to the shorter list length, and applying a rigid-body superposition.

Initial RMSD refers to the C-alpha RMSD between the initial Boltz-2 predicted structure and the UniProt-obtained reference structure used as ground truth. AERIS RMSD refers to the C-alpha RMSD between the AERIS-refined structure, obtained by applying AERIS to the initial Boltz-2 prediction, and the same UniProt reference structure.

Component	Setting
Initial predictor	Boltz-2; method is predictor-agnostic
Attention source	ESM-2_t33_650M_UR50D; method is attention-source-agnostic
Anchor rule	Top 20% by embedding norm and bottom 20% by attention entropy
Smoothing step	C-alpha centroid update with alpha = 0.2
Primary metric	C-alpha RMSD after rigid-body superposition
Main sample size	19 unique UniProt targets

Table 1. Experimental configuration used for the benchmark.

5. Results

Across 19 unique UniProt targets, AERIS reduced mean C-alpha RMSD from 18.07 ± 6.43 Å to 17.09 ± 6.64 Å. The mean absolute reduction was 0.98 Å with a 95% confidence interval of 0.79 to 1.18 Å. The mean percentage reduction was 5.94% with a 95% confidence interval of 4.71% to 7.17%. All 19 unique targets improved. A paired t-test gave $p = 4.13 \times 10^{-9}$, and a one-sided Wilcoxon signed-rank test gave $p = 1.91 \times 10^{-6}$.

Quantity	Value	Unit
Initial RMSD	18.07 ± 6.43	Å
AERIS RMSD	17.09 ± 6.64	Å
Absolute RMSD reduction	0.98 ± 0.41	Å
Percentage RMSD reduction	5.94 ± 2.55	%
Improved targets	19/19	count
Median percentage reduction	6.43	%

Table 2. Main performance summary over 19 unique proteins.

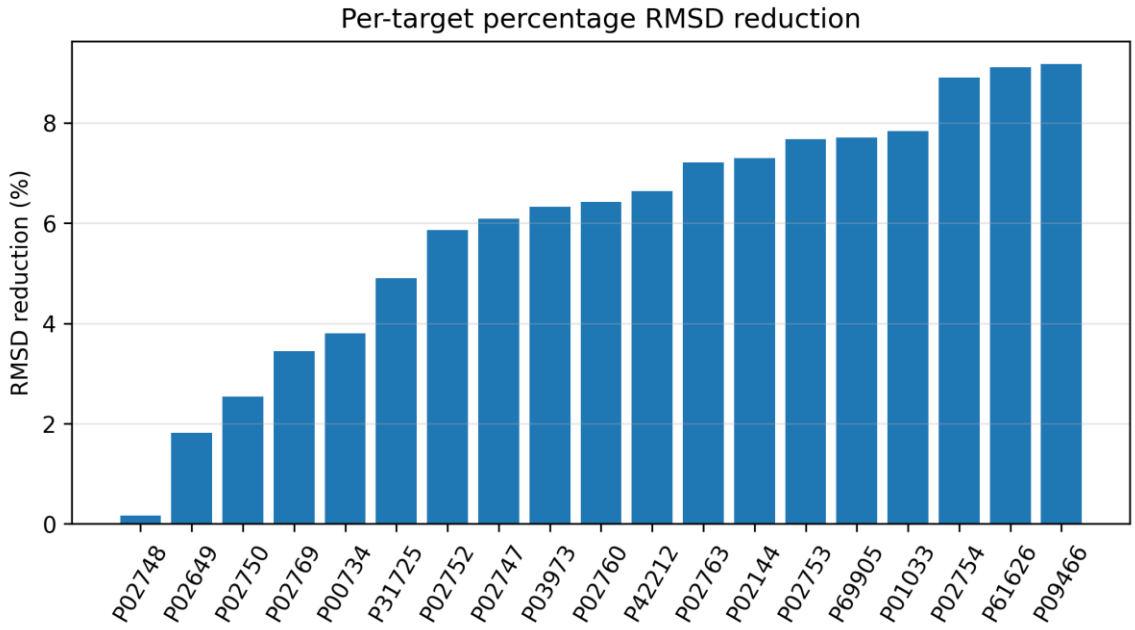

Figure 2. Per-target percentage RMSD reduction, sorted from lowest to highest. The weakest case is P02748, which has a much larger baseline RMSD and only a small absolute reduction.

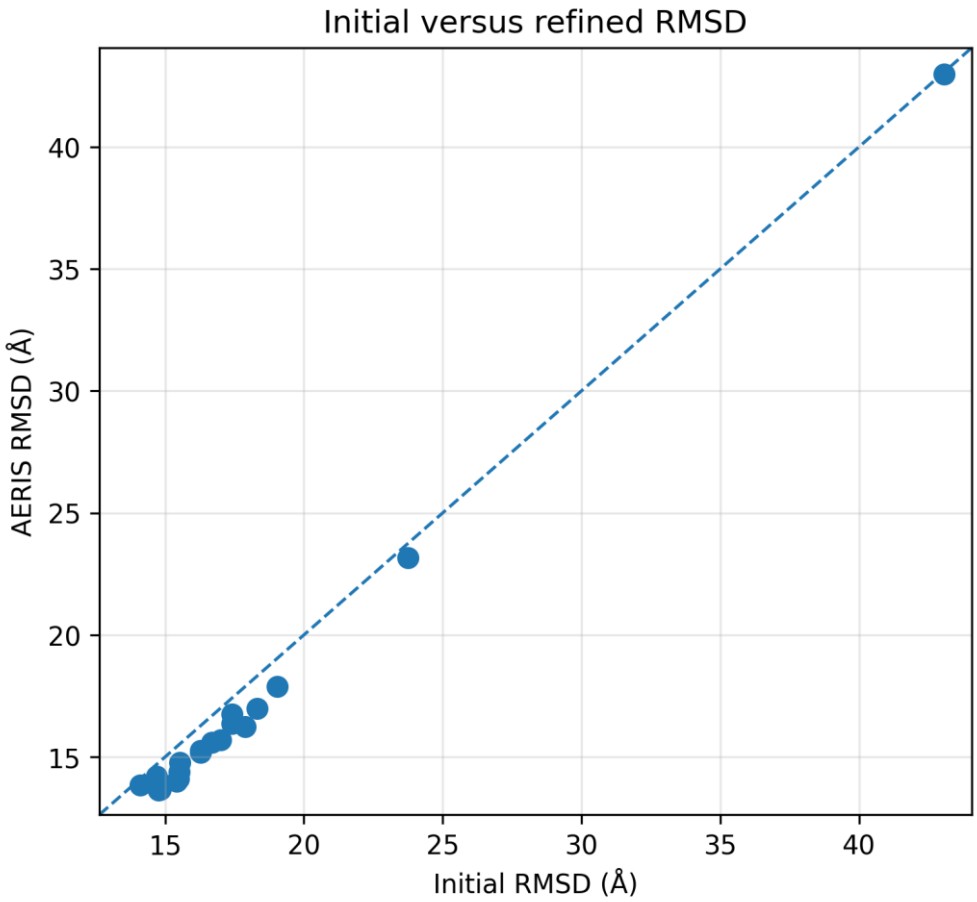

Figure 3. Initial versus refined RMSD. Points below the diagonal indicate improvement after AERIS.

6. Discussion

The current data suggests that attention-entropy-indexed anchors can provide a useful geometric prior for small post-prediction corrections. The most important interpretation is consistency: every unique target improved under the current RMSD calculation. This is stronger than a result driven by one or two large outliers, although the overall benchmark remains small.

The method is intentionally modular. Boltz-2 supplies initial structures in this pilot, but AERIS is not tied to Boltz-2. In principle, the same refinement could be applied to any input structure, including outputs from AlphaFold-like models, RoseTTAFold-like models, other diffusion-based biomolecular predictors, or even experimental models requiring lightweight geometric smoothing. Similarly, ESM-2 is not required by the definition of the method. Any model that supplies residue-residue attention or a comparable residue-wise uncertainty or saliency signal could be substituted.

The current update used in AERIS is global and centroid-based, and future versions could use local neighborhoods, graph smoothing, residue-specific step sizes, or attention-weighted directional updates. These extensions would better connect the attention entropy signal to residue-specific geometry.

7. Limitations

AERIS currently updates only C-alpha coordinates. It does not preserve full backbone geometry or side-chain consistency. Further development can investigate the feasibility of all-atom structural refinement. The benchmark used RMSD, which is useful but can be outlier-sensitive and affected by domain rearrangement and incomplete structures. Future evaluations should include sequence-aligned C-alpha RMSD, lDDT, CAD-score, TM-score, and possibly steric or energy-based quality checks.

8. Conclusion

AERIS provides a lightweight algorithm for attention-entropy-guided protein structure refinement. In the current benchmark, a Boltz-2-initialized and ESM-2-indexed implementation reduced C-alpha RMSD consistently across 19 unique proteins, with a mean improvement of approximately 0.98 Å or 5.94%. The algorithm's main appeal is its replaceable design and lightweight computational cost, as the initial predictor and attention source are not fixed components.

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Appendix A. Full result table

The reviewed result file used for v3 contains 19 unique UniProt targets. The table below reports the exact per-target RMSD values used to compute the main text statistics.

UniProt ID	Initial RMSD	AERIS RMSD	Reduction	Reduction %
P69905	14.77	13.63	1.14	7.71%
P02144	15.50	14.37	1.13	7.29%
P61626	15.41	14.01	1.40	9.11%
P42212	16.27	15.19	1.08	6.64%
P02769	14.70	14.19	0.51	3.45%
P02763	18.31	16.99	1.32	7.21%
P02760	16.68	15.61	1.07	6.43%
P02748	43.05	42.98	0.07	0.16%
P02747	19.04	17.88	1.16	6.09%
P02750	23.76	23.16	0.60	2.54%
P02752	17.40	16.38	1.02	5.86%
P02753	17.00	15.69	1.31	7.68%
P02754	15.49	14.11	1.38	8.91%
P09466	17.88	16.24	1.64	9.18%
P02649	14.10	13.85	0.26	1.82%
P03973	16.29	15.26	1.03	6.32%
P31725	15.53	14.77	0.76	4.91%
P00734	17.40	16.74	0.66	3.80%
P01033	14.82	13.66	1.16	7.83%
Mean	**18.07**	**17.09**	**0.98**	**5.94%**

Table A1. Full RMSD results of all 19 proteins